Input: KS orbitals $\psi_n(\mathbf{r}) = \sum_\mu C_{\mu n}\,\phi_\mu(\mathbf{r})$, energies ε_n, and atom-centered grid points $\{\mathbf{r}_l\}$

1 Fitting coefficients (for every atom P) $O(N)$

$$\sum_{l'} D_{ll'}\, Z_{l'P} = d_{lP}\,, \quad D_{ll'} = \Big[\sum_\mu \phi_\mu(\mathbf{r}_l)\,\phi_\mu(\mathbf{r}_{l'})\Big]^2,$$

$$d_{lP} = \sum_{\mu\nu} \phi_\mu(\mathbf{r}_l)\,\phi_\nu(\mathbf{r}_l)\,(\mu\nu|P)$$

2 Polarizability (for every imaginary time τ) $O(N)$

$$\chi_{PQ}(i\tau) = -2\sum_{ll'} Z_{lP}\, G^{\mathrm{occ}}_{ll'}(i\tau)\, G^{\mathrm{vir}}_{ll'}(i\tau)\, Z_{l'Q}\,,$$

$$G^{\mathrm{occ/vir}}_{ll'}(i\tau) = \sum_{\mu\nu} \phi_\mu(\mathbf{r}_l)\, G^{\mathrm{occ/vir}}_{\mu\nu}(i\tau)\, \phi_\nu(\mathbf{r}_{l'})$$

3 Screened interaction (for every frequency ω) $O(N^3)$

$$\epsilon(i\omega) = 1 - V^{1/2} M^{-1} \chi(i\omega) M^{-1} V^{1/2},$$

$$W^{\mathrm{c}}(i\omega) = V^{1/2}\big[\epsilon^{-1}(i\omega) - 1\big]V^{1/2}$$

4 Self-energy (for every imaginary time τ) $O(N)$

$$\Sigma^{\mathrm{c}}_{\mu\nu}(i\tau) = \sum_{ll'} \phi_\mu(\mathbf{r}_l)\, G_{ll'}(i\tau)\, W^{\mathrm{c}}_{ll'}(i\tau)\, \phi_\nu(\mathbf{r}_{l'})\,,$$

$$W^{\mathrm{c}}_{ll'}(i\tau) = \sum_{PQ} Z_{lP}\,\big[M^{-1} W^{\mathrm{c}}(i\tau) M^{-1}\big]_{PQ}\, Z_{l'Q}$$

5 Quasiparticle energies (analytic continuation) $O(N^3)$

$$\varepsilon_n^{G_0W_0} = \varepsilon_n + \Sigma^{\mathrm{x}}_{nn} + \mathrm{Re}\,\Sigma^{\mathrm{c}}_{nn}\big(\varepsilon_n^{G_0W_0}\big) - v^{\mathrm{xc}}_{nn}$$

sparse grid contraction dense, auxiliary / MO basis